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Enzymes	Mode of action	General scheme of reaction	Example
Lyase	Break chemical bond without addition of water	A – B ⟶ A + B	Decarboxylase
Ligase	Formation of new chemical bonds using ATP as a source of energy	A + B + ATP ⟶ A – B + ADP + Pi	DNA ligase



Telomerase – A Ribonucleo Protein

Telomere protects the end of the chromosome from damage. Telomerase is a ribonucleo protein also called as *terminal transferase*.

8.6.17 Uses of Enzymes

Enzyme	Source	Application
Bacterial protease	Bacillus	Biological detergents
Bacterial glucose isomerase	Bacillus	Fructose syrup manufacture
Fungal lactase	Kluyveromyces	Breaking down of lactose to glucose and galactose
Amylases	Aspergillus	Removal of starch in woven cloth production

8.7 Nucleic Acids

As we know DNA and RNA are the two kinds of nucleic acids. These were originally isolated from cell nucleus. They are present in all known **cells** and **viruses** with special coded genetic programme with detailed and specific instructions for each organism heredity.



Friedrich Miescher *was the first to isolate a non-protein substance in nuclei of pus cells and named it as* '**Nuclein**'.

DNA and RNA are polymers of monomers called **nucleotides**, each of which is composed of a nitrogen base, a pentose sugar and a phosphate. A purine or a pyrimidine and a ribose or deoxyribose sugar is called **nucleoside**. A nitrogenous base is linked to pentose sugar through n-glycosidic linkage and forms a nucleoside. When a phosphate group is attached to a nucleoside it is called a **nucleotide**. The nitrogen base is a